
TCX2005: Information Systems, Management and Organisations

L1: Course Introduction

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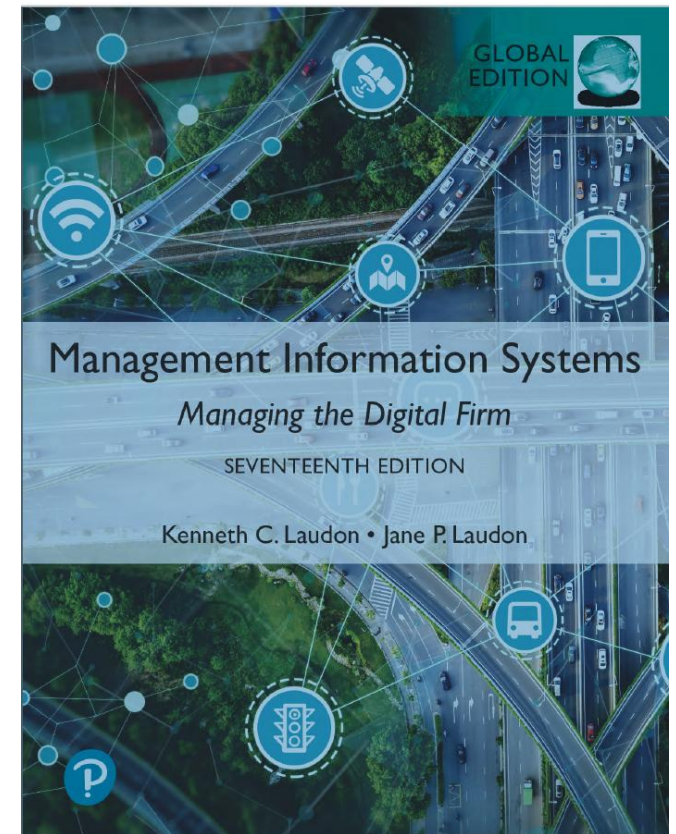
Time & Venues

- Lecture
 - Thursday, 6:45 – 8:45 PM
 - Venue: LT15 (if not zoom)
 - Recordings on Canvas
 - Tutorial (TA: Mr. Yutaro)
 - Starting next week (Week-2)
 - Thursday, 8:45 – 9:45 PM
 - Same as lecture theatre (or same zoom session)
 - Not recorded
 - **Final Exam**
 - Jul 11, 9:00 – 11:00 AM (Exemplify; No internet)
 - Venue: **TBD**
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Learning Resources

- Textbook (recommended)
 - Management Information Systems- Managing the Digital Firm, 17th Ed.,
 - Authors: Laudon, K. C., Laudon, J.P.,
 - Publisher: Pearson
- Other Material
 - Weekly: Reading and Video links shared on Canvas





Lecture Slides

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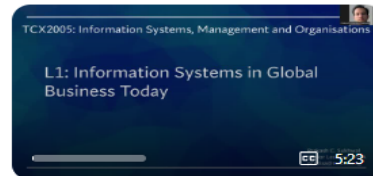
Attendance

Course Readings

↑ L1

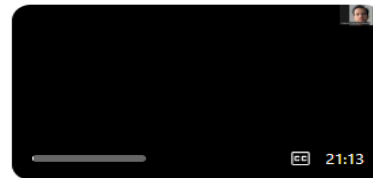
All Dates

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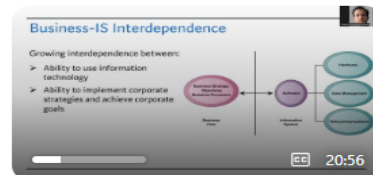
L1_Part-1_Introduction_and_Objectives

• 2 days ago



L1_Part-2

• 2 days ago



L1_Part-3

• 2 days ago



Assessment

Individual (70%)

- **Class Performance: 10%**
 - Tutorial: 7 points (1 point each; T2-T8)
 - Bonus: 3 points (*submit all Tutorials/Activities on time*)
- **Canvas weekly quiz 10%**
 - **Timed Quiz 1 to 6:** Material covered weekly (every 2 sessions)
 - **MUST be done by Canvas deadline (NOT available after deadline)**
- **Final Exam (open book): 50%**
 - In-person; Written (Exemplify (No internet; FULL lock down))
 - MCQs + Short-answer Qs

Group (30%)

- **Team Project: 30%**
 - Proposal: 10% (1 submission per team)
 - Presentation: 20% (*10 mins + 5 mins QA*)

Forming Teams

- T1 in Week-1
- ~5 or 6 Students per team
 - **No individual project**
 - No changes permitted after week-3
- Self sign-up on Canvas:
 - Canvas → People → Project group

Timetable (May 14 – Jul 11, 2026)

Week	Pre-Recorded Lecture (Released every Monday morning)	F2F/Zoom Lecture (Thursday)	Tutorial	Milestones
1		L1: Course Introduction, Information Systems in Global Business Today-I [In-person]	T1: Project Briefing & Team Formation	Teams Ready (Register on Canvas before L3)
2	L2: Information Systems in Global Business Today-II [Pre-recorded]	L3: Information Systems, Organizations, and Strategy-I [In-person]	T2: tutorial (in-class)	
3	Project/Course Consultation (zoom link will be sent)	L4: Information Systems, Organizations, and Strategy-II [Pre-recorded] Project/Course Consultation (before tutorial)	T3: tutorial (zoom)	Project Proposal Due
4	L5: Issues raised by Information Systems [Pre-recorded]	L6: Critical IS Infrastructure – I [Online Via Zoom]	T4: tutorial (zoom)	
5	L7: Critical IS Infrastructure – II [Pre-recorded]	Course and Project Consultation [Online Via Zoom]	T5: tutorial (zoom)	
6	L8: Securing Information Systems: Governance & Risk Management [Pre-recorded]	L9: Enhancing Decision Making [Online Via Zoom]	T6: tutorial (zoom)	Submit Presentation Slides (Deadline: Sunday midnight)
7	L10: Building Information Systems [Pre-recorded]	Online Project Presentation (~3 hrs.)	T7: tutorial (take home)	
8	Course wrap-up and Consultation (zoom link will be sent)	Online Project Presentation (~3 hrs.)	T8: tutorial (take home)	
9	Final Exam (Jul 11, 2026) [Written; in-person]			Final Exam

F2F/Online Classes

Lectures

- L1 (F2F; Today)
- L3 (F2F; May 21)
- L4 (online; May 28 [for consultation](#))
- L6 (online; Jun 4)
- L7 (online; Jun 11 [for consultation](#))
- L9 (online; Jun 18)
- [Project Presentation-I](#) (Jun 25)
- [Project Presentation-II](#) (Jul 2)

TCX2005 > Zoom

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Start Time

Topic

Thu, May 28 (Recurring)
12:00 AM

[Weekly Lecture](#)

Thu, Jun 4 (Recurring)
12:00 AM

[Weekly Lecture](#)

Thu, Jun 11 (Recurring)
12:00 AM

[Weekly Lecture](#)

Thu, Jun 18 (Recurring)
12:00 AM

[Weekly Lecture](#)

Thu, Jun 25 (Recurring)
12:00 AM

[Weekly Lecture](#)

Thu, Jul 2 (Recurring)
12:00 AM

[Weekly Lecture](#)

F2F/Online Tutorials

Tutorials

- T1 (Today: Project Briefing)
- T2 (F2F; May 21)
- T3 (online; May 28)
- T4 (online; Jun 4)
- T5 (online; Jun 11)
- T6 (online; Jun 18)
- T7 (Jun 25; Take-home)
- T8 (Jul 2; Take-home)

TCX2005 > Zoom

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Thu, Jun 25 (Recurring) 12:00 AM	Weekly Lecture
Thu, Jul 2 (Recurring) 12:00 AM	Weekly Lecture

Communication

- Canvas
 - Announcement, Files (lecture notes, reading materials, etc.)
- Email
 - Put “TCX2005” in the subject line
 - **DO NOT** email directly through Canvas Inbox
- **Important**
 - Check Canvas and your NUSNET email regularly for announcements and updates



Project Briefing



What is the Project about?



Goal & Deliverables



Submissions & Evaluations



Timelines

Theme: Digital Transformation at Work

- Digital Transformation is the integration of digital technology into all areas of a business or organization, fundamentally changing how you operate and deliver value.
 - It is not just about technology—it requires cultural change, process redesign, and new ways of thinking.
 - During TCX2005 we will cover aspects of digital transformation under:
 - IS in global business: L1/L2
 - The underlying strategy: L3/L4
 - Governance & Risk: L8
 - Decision Making & Infrastructure: L6/7/9
-

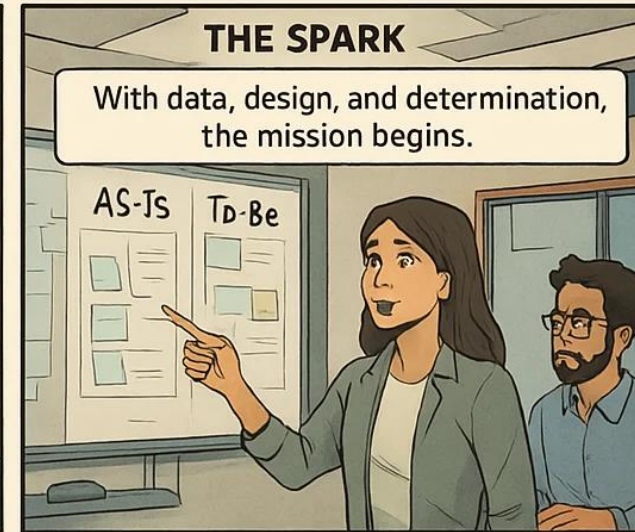
Theme: Digital Transformation at Work

Examples:

- **Operational Reporting:** Automate daily/weekly/monthly reporting from multiple inventory houses, production facilities, or regions using cloud dashboards or reporting tools. This reduces repetitive manual data entry and frees up hours of valuable time.
- **Leave Application Process:** Replace paper/email-based forms with an automated HR portal or chatbot.
- **Expense Claims:** Scan receipts into a system that auto-generates reports instead of Excel + email workflows.
- **Inventory Tracking:** A mobile app for checking stock levels rather than pen-and-paper logs.

These are small, common processes, which after automation, can save time, reduce errors, and free up staff for more meaningful work.

TRANSFORMERS OF TOMORROW: FROM CHAOS to CLARITY



Have fun while you transform!!

Project Overview

Goal: Identify a manual process in your workplace and redesign it using digital technologies.

Focus Areas

1. Understand current process (As-Is)
 2. Map user journey and pain points (from point 1)
 3. Propose a digitally transformed process (To-Be)
 4. Highlight impact, tech enablers, constraints & requirements
-

Project Timeline and Key Dates



Team formation via Canvas

Students self-register their team using the Canvas platform



Proposal due (submit on Canvas)

Team leader submits the proposal through Canvas



Submit slides (Sunday 23:59)

Presentation slides are due by Sunday midnight



Online (Zoom) Project Presentations (~3-4 hrs.)

Whole Team presents projects online
Duration: 10 minutes + 5 minutes Q&A (per team)

Team Formation

- 5-6 members per team
 - **No individual** projects allowed
 - Finalize teams by Week 2 on Canvas:
 - Go to Canvas → People → Project Group
 - No changes after Week 3
-

Suggested Workflow

- **Identify** a process relevant to your current or past workplace
 - **Map** the As-Is process using swimlanes or flowcharts
 - **Capture** the user journey and **key** stakeholders
 - **Design** To-Be process powered by digital tools (AI, apps, other IS systems, etc.)
 - **Specify** constraints & requirements (budget, tech, change mgmt.)
 - **Comment** on possible impact of the proposed solution
-

Assessment Breakdown

- Proposal 10 %
 - One submission per team on Canvas
 - Format: 2-slides (cover and references not counted)
 - Due: Week 3 (See timetable)
 - Presentation 20 %
 - 10 mins + 5 mins Q&A per team
 - Format: Up to 8 Slides (cover and references not counted)
 - Conducted in Week 7 or 8 (Online)
-

Example: 2-Slide Proposal Template

Title of the Project

Team Members

S1

Brief Description of
Current Process

Key Stakeholders / Users

S2

Motivation for Digital
Transformation

Initial Ideas for To-Be Process

Anticipated Tech / Tools

Digital Transformation Proposal Overview

Project Title & Team Members

Introduction of project and key personnel



Motivation for Transformation

Reasons driving the change



Initial To-Be Process Ideas

Preliminary concepts for new processes



Current Process Description

Overview of existing procedures



Key Stakeholders/Users

Identification of main participants



Anticipated Tech/Tools

Expected technologies and tools



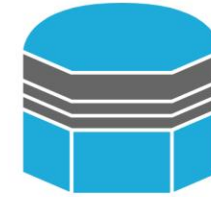
Final Presentation Guidelines

Project Evaluation Breakdown

- **Duration:** 10 minutes + 5 minutes Q&A
- **Format:** Online via Zoom
- **Required:** Visuals (process diagrams, flowcharts, etc.)
- **MUST Submit Slides by:**
Week 6 Sunday midnight

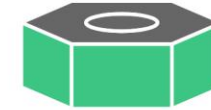
Problem Understanding & As-Is Analysis

Clear explanation and visualization of current processes



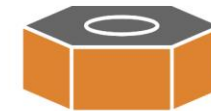
To-Be Process Design

Creativity and feasibility of redesigned processes



Constraints, Requirements & Change Management

Recognition of limitations and management



Stakeholder & User Journey Mapping

Identification of key users and their needs



Value Addition & Impact

Savings in time and reduction of errors



Presentation Quality & Teamwork

Clarity and engagement in delivery

Example

Problem Context

Organizational Baseline



Organization: Company S
(Major Enterprise)

Process: Corporate Financial
Expense Processing

Scale: Average 1,450 cases per
month

Start Boundary: Employee incurs
expense & submits proof.

End Boundary: Approved data
registered in Financial ERP (SAP)
& employee reimbursed.

Key Stakeholders

**Card Users
(Employees)**

Submit
physical/digital
evidence.

**Department
Managers
(Approvers)**

First-line business
justification.

**Finance
Reviewers**

Final verification
against
compliance rules.

**IT
Administrators**

Maintain legacy
financial ERP.

Current Inefficiency Drivers



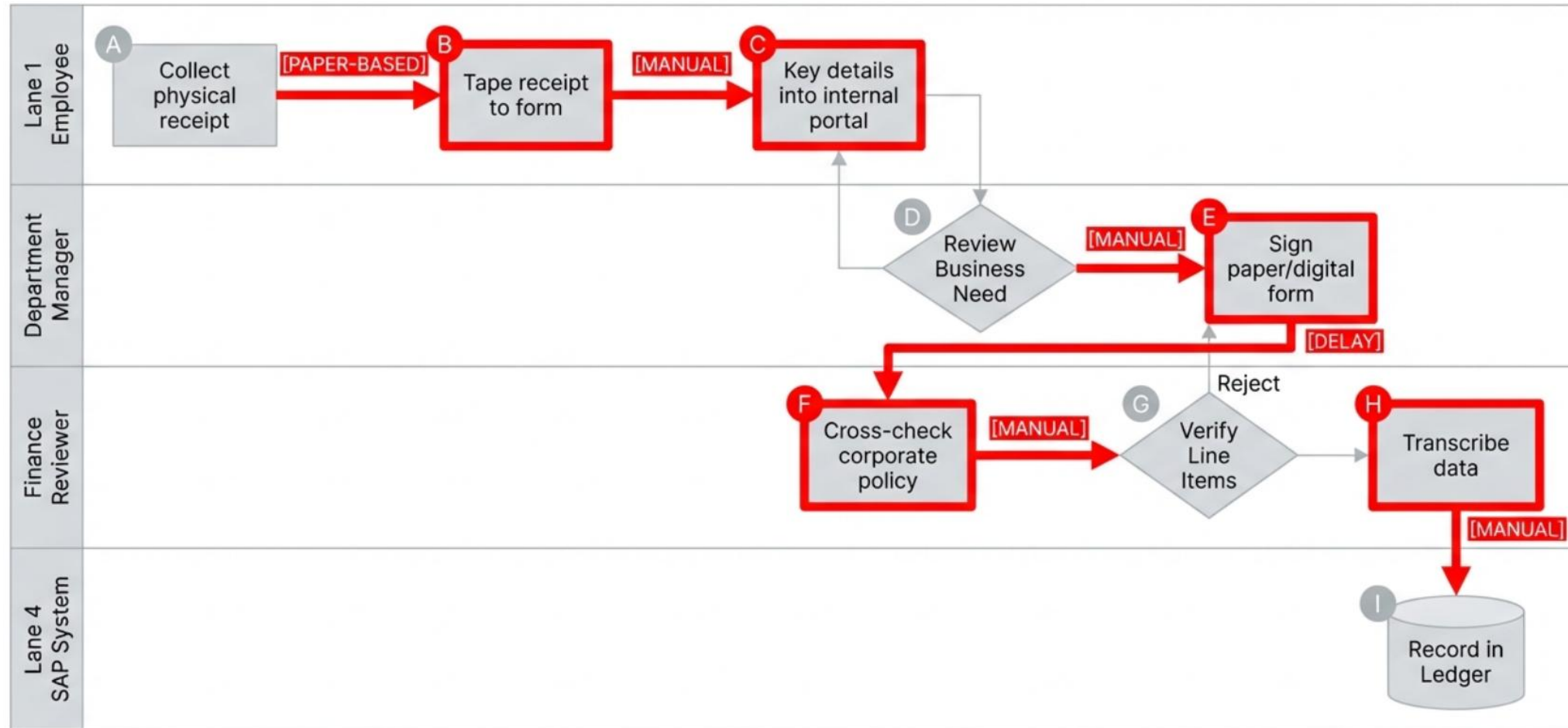
Process relies strictly on:

- Physical/paper receipts
- Manual ERP data entry
- Human verification of complex corporate compliance rules

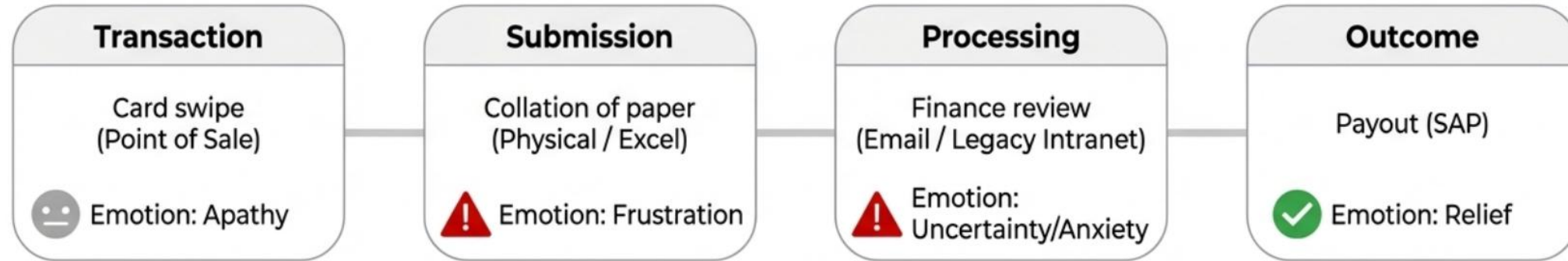
Resulting Pathology:

- High error rates (typos, omissions, policy violations)
- Severe processing bottlenecks (averaging 24+ hours per batch)
- Massive resource drain on the finance team for rule-based verification

AS-IS Process Flow

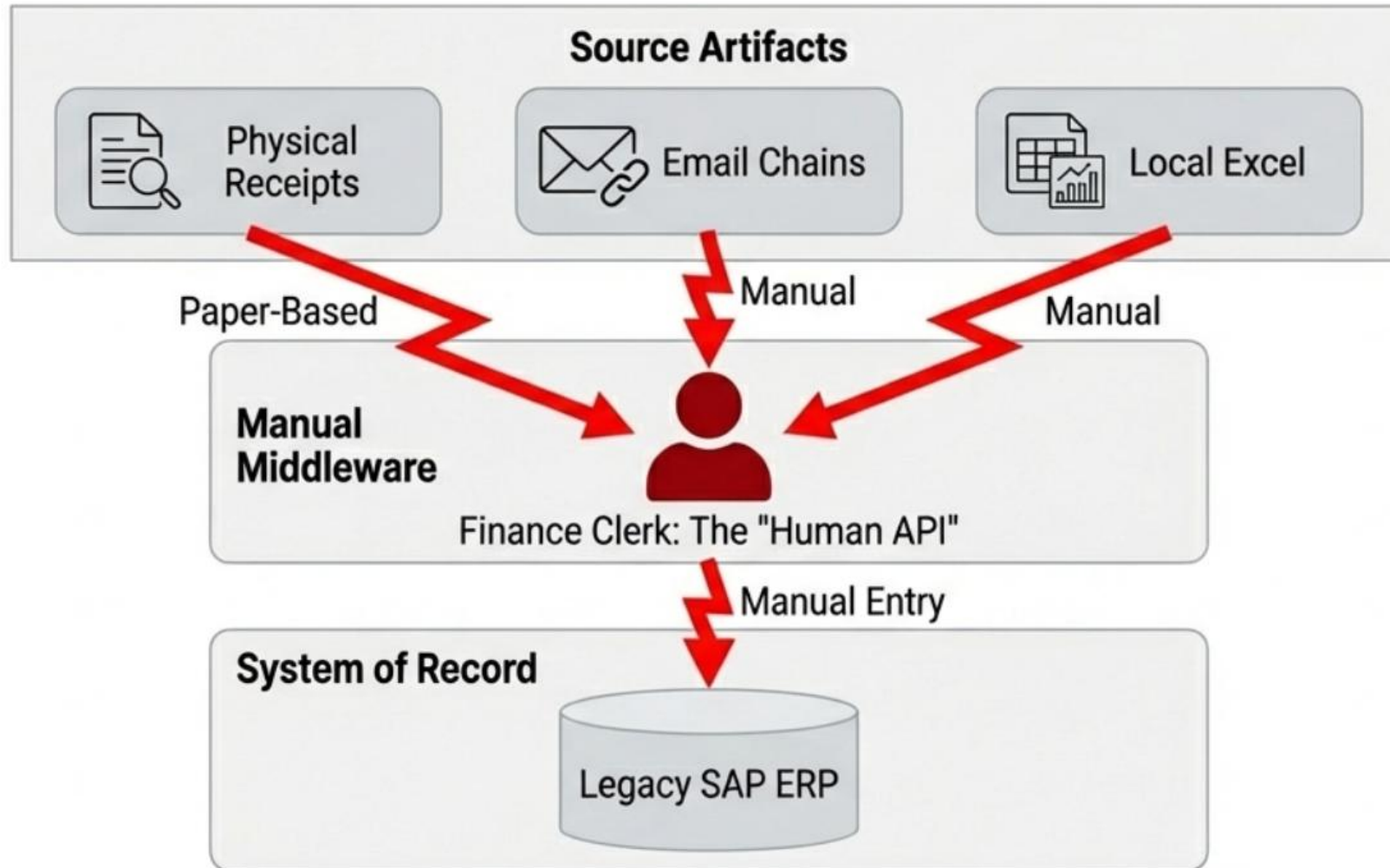


AS-IS User Journey & Pain Points

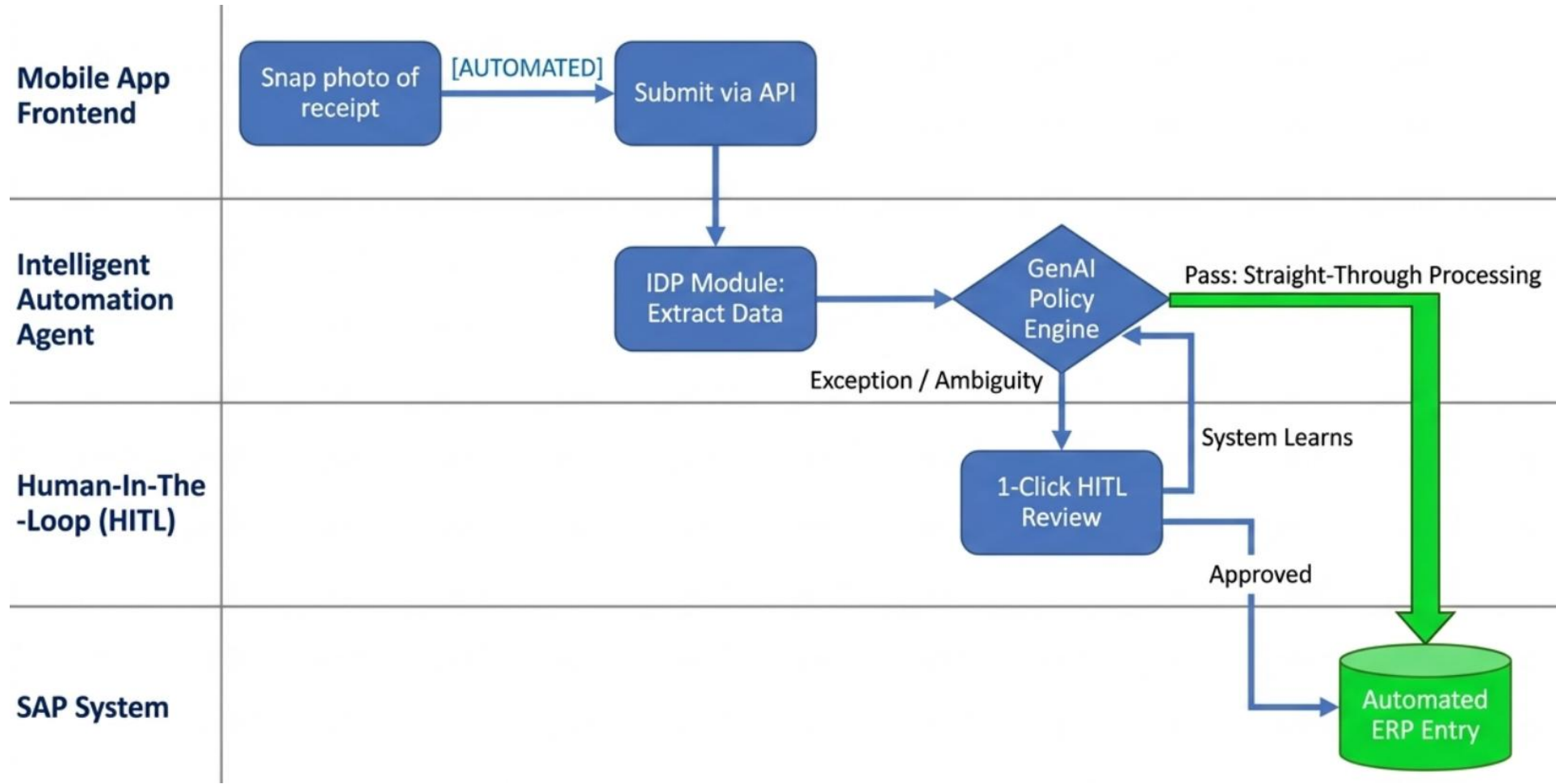


Process Step	Pain Point	Severity	Frequency
Receipt Submission	Lost physical receipts; manual scanning and taping to forms.	H	H
Policy Check	Finance manually checks compliance rules per line item against complex corporate policy documents.	H	H
Exception Handling	End-to-end delays caused by looping email chains for missing information.	H	M
ERP Data Entry	Typos, omissions, and manual keying errors causing reconciliation failures.	M	H

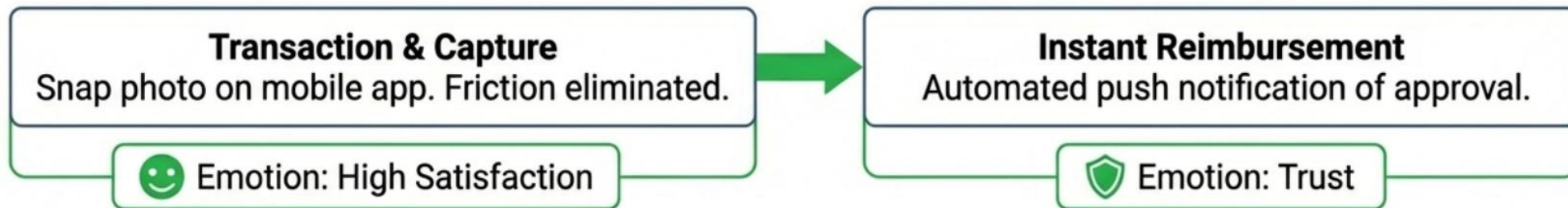
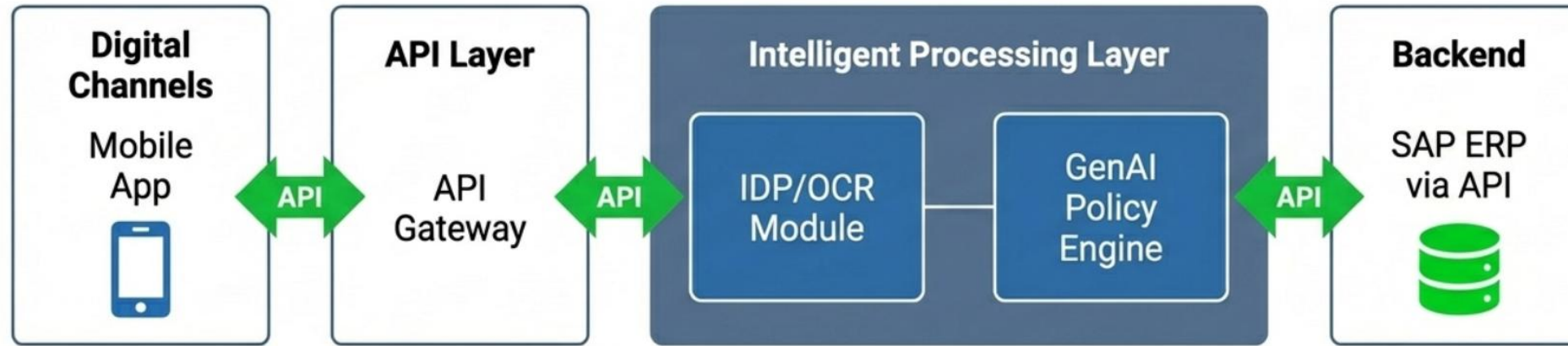
AS-IS System Architecture



TO-BE Process Flow



TO-BE System Architecture & Planned User Journey



Clear Traceability

Pain Point	Capability Gap	TO-BE Solution	Measurable KPI Impact
Manual transcription of physical paper	Unstructured data ingestion	IDP/OCR extraction module	Data entry time reduced to 0 seconds .
Slow, manual corporate compliance checks	Automated decision engine	Generative AI Policy Engine	Verification time reduced from minutes to <2 seconds per receipt.
Reimbursement status is a black box	Real-time tracking architecture	Mobile Frontend & API Gateway	Cycle times cut from weeks to days .
Errors and typos in SAP ledger entry	System integration (STP)	Automation Orchestrator / API	99.9% error reduction in financial system-of-record.

Support and Resources

- Project-related consultations via Zoom (Week 3 and 5; see timetable)
 - Use GenAI responsibly (e.g., visual aids, documentation support)
-

References

- Laudon, K. C., & Laudon, J. P. (2021). Management information systems: Managing the digital firm (17th ed.). Pearson.
 - Icons: <https://www.flaticon.com/>
 - Some visuals in the course slides are generated using AI tools: [NotebookLM](#), [NapkinAI](#)
-